PHARMACEUTICAL DEVELOPMENT AND TECHNOLOGY
2023, VOL. 28, NOS. 3–4, 318–332
https://doi.org/10.1080/10837450.2023.2194989

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REVIEW ARTICLE

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The electronic tongue: an advanced taste-sensing multichannel sensory tool with global selectivity for application in the pharmaceutical and food industry

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ABSTRACT

Taste is a crucial organoleptic characteristic that determines whether a substance will be accepted for delivery through the mouth. However, the vast majority of medications have an unpleasant taste. Drugs with a bitter taste are often depicted using a variety of flavouring compounds to increase patient acceptance and compliance. Human panellists are the principal means of assessing the flavour of medication ingredients and formulations. Due to the toxicity of medications and subjectivity of taste panellists, as well as issues with hiring taste panellists and panel upkeep when working with unpleasant items, the use of sensory panellists in the industry is particularly challenging and troublesome. Furthermore, tests cannot be conducted on compounds that have not received FDA approval. As a result, the analytical taste-sensing multichannel sensory system known as the electronic tongue (also known as the artificial tongue or e-tongue) helps in reducing the number of samples that are ought to be assessed by trained sensory panels and also when the sample to be tasted is injurious or harmful for the concerned person. Therefore, e-tongue has advantages like lowering reliance on human panels. The working theory, the sensors used, and the pharmaceutical and food applications are covered, along with the major software used, difficulties, and future scope are also highlighted.

ARTICLE HISTORY
Received 6 January 2023
Accepted 21 March 2023

KEYWORDS
Electronic tongue; sensor arrays; chemometric techniques; pattern recognition

1. Introduction

Nerves in the brain convert taste sensations into electric signals physiologically. The perception and identification of taste quality are based on the brain's creation or recognition of activated sensory nerve patterns. Evaluation of taste masking efficiency is done using a variety of techniques, including the human panel method; the assessment of frog flavour nerve responses, and the electronic tongue (ET). The human panel method is a technique of analyzing bitter taste using a traditional human taste panel but this approach is difficult because of the expense, time requirements, and safety concerns involved (Figure 1(A,B)). In the assessment of frog flavours nerve response, *frog taste* cell *responses* are modulated by activities of parasympathetic postganglionic efferent fibers innervating the cells. The advantage of using the frog taste nerve response method is that it can determine the bitter or sharp taste perception of vertebrates (Woertz et al. 2011; Lu et al. 2022).

The third technique ET is an analytical device mainly used to classify the tastes of various chemical substances in liquid phase samples which offers a quick, simple, and accurate evaluation of a variety of oral formulations including chewable tablets, quickly dissolving tablets and films, oral dispersive lozenges, sublingual tablets; as well as some nasal delivery products which are highly correlated with organoleptic taste (Braz et al. 2022). It could discriminate between chemicals that have distinct taste modalities as well as between drugs that produce the same basic flavour. It has also been demonstrated to be able to distinguish between formulations with various amounts of sugar or flavour in a way that is consistent with their effectiveness at concealing tastes as evaluated by a human taste panel (Łabańska et al. 2019).

It is a system that usually consists of an array of non-specific chemical sensors combined with appropriate data acquisition systems and chemometric tools. During sample assessment, the ET sensor array produces an unresolved analytical signal, which is correlated with the chemical composition of the sample. This relies on the proper choice of the sensor array, comprising sensors with pronounced sensitivity towards the substances and properties of interest. The resulting signal can be processed by various multivariate data analysis techniques to extract quantitative and qualitative information about the sample. Principal component analysis (PCA), discriminant factorial analysis (DFA), cluster analysis, and partial least squares (PLS) are among the chemometric methods used to describe the relationship between the sensors and test preparations (Kirsanov et al. 2019).

PCA is a technique used to emphasize variation and bring out strong patterns in a dataset. It is often used to make data comparatively easier to explore and visualize. **DFA** is used to determine which variables discriminate between two or more naturally occurring groups. Cluster Analysis is the process to find similar groups of objects in order to form clusters. It is an unsupervised machine learning-based algorithm that acts on unlabelled data. A group of data points would comprise together to form a cluster in which all the objects would belong to the same group. **PLS** is a comparatively quicker, efficient, and optimal regression method based on covariance. It is recommended in cases of regression where the number of explanatory variables is high, and where it is likely that there is multicollinearity among the variables, i.e. that the explanatory variables are correlated. It is the most widely

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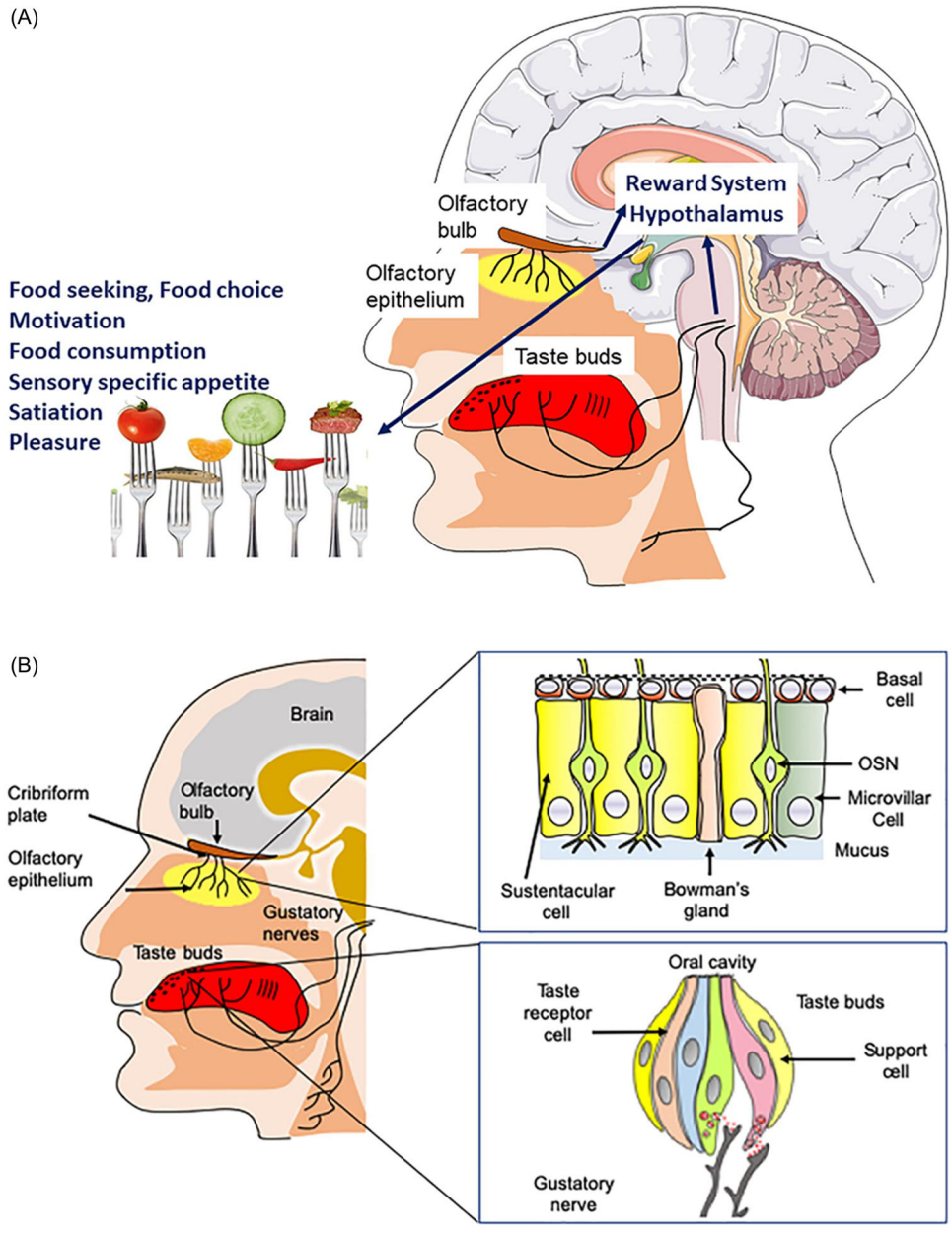


Figure 1. (A) Hedonic sensory signal and food intake. Signals coming from the chemosensory systems (olfaction, taste, and chemesthesis) directly inform the brain of the sensory quality of food. The main brain areas involved are those of the reward system as well as hypothalamic nuclei. These will be involved in different components of feeding behavior such as motivation and pleasure to eat but also sensory-specific appetite and satiation. (COVID-19-Induced Smell and Taste Impairments: Putative Impact on Physiology© 2021 Meunier, Briand, Jacquin-Piques, Brondel, and Pénicaud, Used under Creative Commons Attribution License (CC BY-NC-SA 2.0). (B) Anatomy of taste and olfaction. Volatile odorant molecules are detected by olfactory sensory neurons (OSNs) located in the olfactory epithelium. OSNs are neurons extending ciliated dendrites in the periphery and with axons forming synapses with glomeruli in the olfactory bulbs situated in the brain. Taste buds located on the tongue are involved in the detection of soluble tasting compounds. The detection is achieved by taste receptor cells located in taste buds. Taste information is transmitted to the brain by three nerves, including chorda tympani, the glossopharyngeal, and the vagus nerves. (COVID-19-Induced Smell and Taste Impairments: Putative Impact on Physiology© 2021 Meunier, Briand, Jacquin-Piques, Brondel, and Pénicaud, Used under Creative Commons Attribution License (CC BY-NC-SA 2.0).

used method to obtain calibration models for numerical predictions of various quality parameters (Kirsanov et al. 2019; Lozano-Torres et al. 2022).

For data processing, the signal acquisition for ET is performed by the elto data logger, which automatically tracks and records environmental parameters over time, enabling measurements, documentation, analysis, and validation of parameters. It has a sensor that collects data, after which statistical software like SPSS will analyze the information. There are several statistical software that has been applied, depending on the research group, which goes from commercial ones like SPSS (Statistical package for the social sciences), SAS (Statistical analysis system), STATISTICA, etc to open access software like R and PSPP (Wang et al. 2022).

These software's assists the user in describing data and looking for a correlation or relationship between one or more variables and provides a graphical point-and-click user interface for non-technical users. They use a suitable method of pattern recognition and multivariate calibration. The sensor data is converted into taste patterns. As a result, the taste and flavour components of the human palate can be detected. Additionally, it can assess the quantitative content and distinguish between different types of tastes or flavours. They are commonly used in food analysis and

the pharmaceutical industry, to classify and analyze the tastes of multicomponent combinations (Paup et al. 2019; Gharibzahedi et al. 2022). It has several benefits, including quick analysis times, cheap cost per analysis, ease of construction, and non-destructive nature. It can also be automated and miniaturized for real-time analysis (Ghasemi-Varnamkhasti et al. 2018).

2. Working principle of the e-tongue

ET consists of three elements; The sample-dispensing chamber or automatic sample dispenser; an array of sensors of different selectivities; and software for data processing (Figure 2). The sample-dispensing chamber is a device that comprises an upper hopper having an upper opening for receiving medicament and a lower opening for dispensing. The array of sensor senses the sample and convert that into signals; a signal pattern is generated, which can be correlated to certain features or qualities of the sample. Software like SPSS and SAS is used to process the data for an image recognition system that mimics brain functions. The analysis of liquid phase samples is performed directly; solid phase samples have to be dissolved prior to analysis (Podrażka et al. 2017).

Almost all electronic tongue or taste sensors developed are based on potentiometry or voltammetry. Potentiometry is a quantitative technique used in analytical chemistry to find the concentration of a solute in a solution. It is based on the difference in the potential between two electrodes which is measured using a high-impedance voltmeter. On the other hand, Voltammetry is the study of the current response of a chemical under an applied potential difference (Cuartero et al. 2022).

3. Sensors used in e-tongue

A systematic description of the type of sensors used for sensing the taste of different types of samples in e-tongue are illustrated in Table 1.

3.1. Potentiometric sensors

These measure the potential difference between two electrodes under the conditions of no current flow. The measured potential may then be used to determine the analytical quantity of interest, generally the concentration of some component of the solution (Munekata et al. 2023). From the produced membrane, a disc with a diameter of 10 mm was punched. It was then attached using Polyvinyl chloride/Tetrahydrofuran adhesive to a polymeric ring support with an internal diameter of 8 mm and installed in an electrode body for potentiometric measurements (Figure 3(A)) (Cho and Moazzem 2022). All tests were performed using a cell of the following type at room temperature:

Ag|AgCl|3 M KCl sample solution| membrane |inner filling solution (0.01 M NaCl; AgCl; Ag.

Potentiometric sensors may be used to build a number of electronic tongues for a range of uses, including food identification, environmental monitoring, and medical diagnostics. Potentiometric ETs have been used to examine a variety of parameters of milk and dairy product quality. For instance, they have been employed to differentiate and categorize milk based on its fat content and to classify natural, fermented, and ultra-high temperature milk using potentiometric fingerprints obtained from the 16 sensors using PCA (Tazi et al. 2018); to analyze the organoleptic characteristics (Minghui et al. 2019); to identify taste alterations in goat and cow milk (Tazi et al. 2017); in the detection of spoilage and the identification of goat milk adulterations with cow milk (Dias et al. 2009; Poghossian et al. 2019) or detection of water added to dairy products among other things (Mabrook et al. 2006).

A potentiometric electronic tongue made of lipid polymeric sensor membranes, as demonstrated by Ghrissi et al. in 2021, can be used as a practical, non-invasive, and direct tool to distinguish between and quantify four common bacteria (*E. coli*, *E. faecalis*, *S. aureus*, and *P. aeruginosa*) in aqueous solutions that are known to be food or water contaminant bacteria. The known ability of the lipid sensor membranes to form electrostatic interactions or hydrogen bonds with hydroxyl, carbonyl, and/or amine groups that can be found, in varying numbers, in the outer membranes of the referred bacteria, could be a possible explanation for the sensor device's satisfactory qualitative and quantitative predictive performances.

3.2. Voltametric sensors

This evaluates the current reaction as a function of the applied potential. A voltammogram, which measures the current as a function of the cell potential I = f(E), is produced by changing the potential either continuously or stepwise. It normally has three

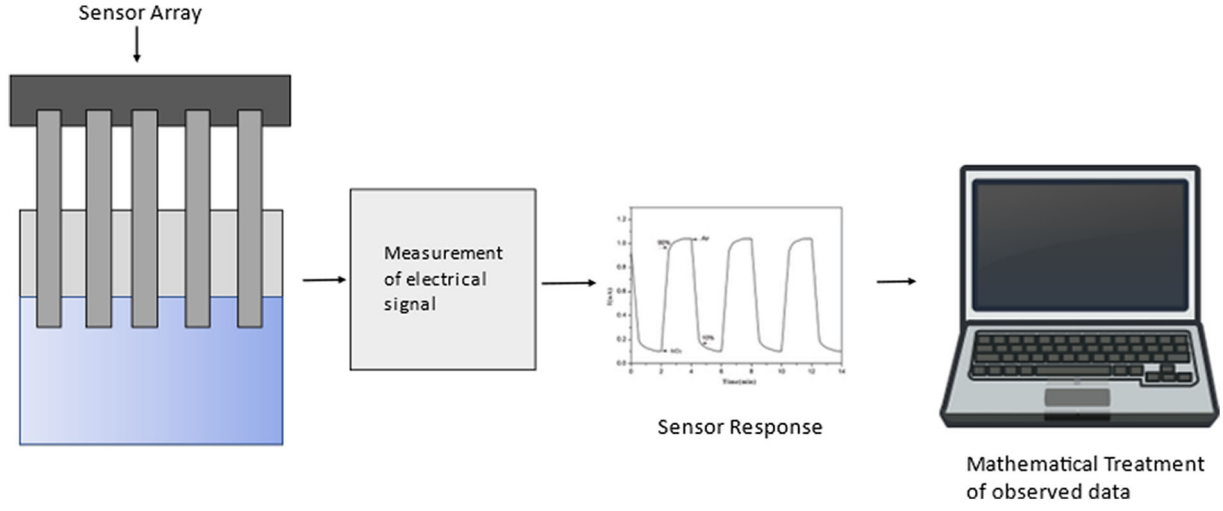


Figure 2. Diagrammatic representation of the working of the Electronic Tongue in which the array of sensor senses the sample and convert that into signals; a signal pattern is generated, which can be correlated to certain features or qualities of the sample.

Table 1. A systematic description of the type of sensors used for sensing the taste of different types of samples in E-tongue.

E-tongue basis	Sensor details	Number of sensors	Application/type of sample sensed	Chemometric Techniques	Reference
1) Potentiometric based E-Tongue	Alpha MOS αAstree I	7	Sensing taste changes in apricots during storage.	CDA and PLS regression	Kantor et al. (2008)
2) Bioelectronic tongue	Bioelectronic tongue including three enzymatic biosensors based on tyrosinase and phthalocyanines as electron mediators	3	Monitoring the aging of beers	PCA, LDA	Ghasemi-Varnamkhasti et al. (2012)
3) Potentiometric e-tongue, Ion- sensitive field- effect transistor (ISFET) based E-Tongue	α-Astree 2 e-tongue (Alpha M.O.S., Toulouse, France)	7	Assessment of taste change in irradiated raw fish	PCA	Feng et al. (2017)
4) A chemically modified field-effect transistor (FET) based E-Tongue	α-Astree e-tongue (Alpha M.O.S., Toulouse, France)	7	Determination of fish freshness	LDA, PLS	Han et al. (2015)
5) Pulse voltammetric e-tongue	The designed system includes 8 metallic working electrodes grouped in two e-tongues	8	Shelf-life assessment of fresh cod in cold storage	PCA, PLS	Ruiz-Rico et al. (2013)
6) Voltammetric electronic tongue	The V e-tongue comprised 4 working electrodes: gold, silver, platinum, and palladium electrode	4	Monitoring of quality and storage time of unsealed pasteurized milk	PCA, Cluster analysis, PLSR	Wei et al. (2013)

CDA: Critical discourse analysis; PLS: Partial least squares; PCA: Principal component analysis; LDA: Linear discriminant analysis: PLSR- Partial least squares regression.

electrodes: the working electrode, the auxiliary electrode, and the reference electrode (Figure 3(B)) (González-Calabuig and del Valle 2018).

3.2.1. Working electrode

The working electrode is made of an inert substance like gold, platinum, glassy carbon, iridium, or rhodium, among others. When this occurs, the working electrode acts as the surface on which the electrochemical reaction occurs. sites where redox reactions take to happen. The surface area should be extremely small (a few mm^2) to restrict current flow (Gutes et al. 2006).

3.2.2. Reference electrode

The working electrode potential is determined using an Ag/AgCl reference electrode. As long as no current passes through a reference electrode, it ought to have a constant electrochemical potential (Parra et al. 2006).

3.2.3. Auxiliary electrode

A conductor that completes the cell circuit is a stainless-steel counter electrode. The conductor is often inert. The counter electrode is where the current flows out of the solution after entering through the working electrode. In the redox process, it plays no part (Cetó and Valle, 2022).

The exact application of potentials and the measurement of current are made possible by the employment of the three electrodes. The working electrode's potential is measured and stabilised using the reference electrode as a standard. The reference electrode is subjected to little or no current to achieve this. Thus, the working electrode receives all current flowing through the auxiliary electrode (Blandon-Naranjo et al. 2022). The three sensors recorded voltammograms are combined into a data matrix. The matrix is then subjected to a chemometric method to get either qualitative or quantitative data. Principal component analysis (PCA), a pattern recognition technique, may be used to find collections of data having similar characteristics (clusters). Identifying unknown samples may be done using the PCA model if it can be used to determine the origin of known samples. When it comes to quantitative data, a calibration model is created using a partial least squares (PLS) approach using a training set of samples with predetermined values for the attributes that need to be identified. If the calibration model's predictions are accurate, it may then be applied to the voltammograms that the tongue recorded to anticipate the needed attributes in unidentified samples. The benefits of voltammetric electronic tongues over potentiometric devices include the greater information obtained (a full voltammogram as opposed to a single potential per sensor) and the higher sensitivity (especially when electrochemical preconcentration is possible), which allow one to take into account not only major components of the sample but also trace substances that can be extremely informative about the origin and quality of food products. The primary disadvantages of voltammetric devices, however, are the more complex experimental designs and, particularly, the datasets, which frequently call for very advanced chemometric models for data processing (González-Calabuig and del Valle 2018).

Arrieta et al. (2010) use polypyrrole polymerized onto Pt disks and doped with different modifiers as working electrodes using partial least squares (PLS) for the prediction of bitterness and alcoholic strength in beers.

De Sá et al. (2016) use Glassy carbon electrodes modified with multi-walled carbon nanotubes containing metal (Pd, Au, Cu, Ni, Co) as working electrodes using an artificial neural network (ANN) for determination of galactose, glucose, xylose, and mannose in sugar cane bagasse.

González-Calabuig and del Valle (2018) use Six graphite–epoxy modified composite electrodes using an artificial neural network (ANN) for the determination of ethyl phenol metabolites in wines.

3.3. Conductometric sensors

When Contacting Sensors are used, the conductivity is measured by applying an alternating electrical current to the sensor electrodes (that together make up the cell constant) immersed in a solution and measuring the resulting voltage. The solution acts as the electrical conductor between the sensor electrodes (Figure 3C) (Łabańska et al. 2019). The potential is varied continuously to determine the current as a function of the cell's potential,

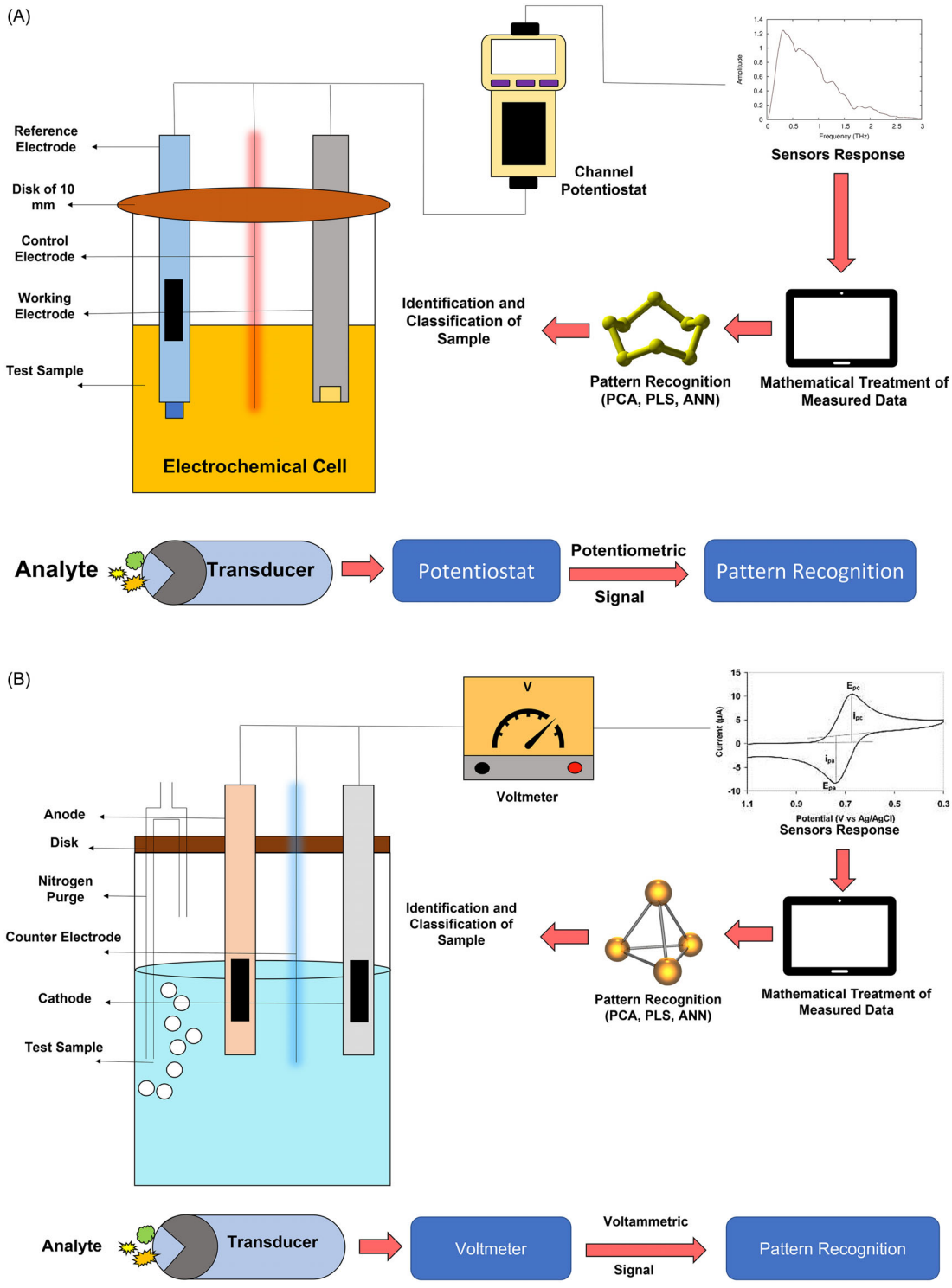


Figure 3. (A) Diagrammatic representation of the working of potentiometric sensors. The apparatus consists of three electrodes; namely, the reference electrode, auxiliary electrode, and working electrode; these electrodes are further connected to channel potentiostat and are dipped into the test sample; channel potentiostat provides the sensor response which provides the pattern recognition (PCA, PLS, ANN techniques) that ultimately leads to the identification and classification of the test sample. (The graph has been reprinted from THz pulse spectrum© 2007 Han-Kwang, Used under Creative Commons Attribution License (CC BY-NC-SA 2.5). (B) Diagrammatic representation of the working of voltammetric sensors. The apparatus consists of three electrodes; namely, anode, cathode, and counter electrode; these electrodes are further connected to the voltmeter and are dipped into the test sample; voltmeter provides the sensor response which provides the pattern recognition (PCA, PLS, ANN techniques) that ultimately leads to the identification and classification of the test sample. (The graph has been reprinted from cyclovoltammogram© 2005 Lifer21, Used under Creative Commons Attribution License (CC BY-NC-SA 2.0). (C) Diagrammatic representation of the working of conductometric sensors. The apparatus consists of three electrodes; namely, anode, cathode, and counter electrode; these electrodes are further connected to the conductometer and are dipped into the test sample; conductometer provides the sensor response which provides the pattern recognition (PCA, PLS, ANN techniques) that ultimately leads to the identification and classification of the test sample. (The graph has been reprinted from *Development of Conductometric Sensor Based on 25,27-Di-(5-thio-octyloxy)calix[4] arene-crown-6 for Determination of Ammonium*© 2016 Saiapina, O. Y.; Kharchenko, S. G.; Vishnevskii, S. G.; Pyeshkova, V. M.; Kalchenko, V. I.; Dzyadevych, S. V, Used under Creative Commons Attribution License (CC BY-NC-SA 2.0). (D) Diagrammatic representation of the working of amperometric sensors. The apparatus consists of three electrodes; namely, anode, cathode, and counter electrode; the cathode and anode are connected to the voltmeter while the counter electrode is connected to the amperemeter, all these electrodes are further connected to the voltmeter and are dipped into test sample; amperemeter and voltmeter together provide the sensor response which provides the pattern recognition (PCA, PLS, ANN techniques) that ultimately leads to the identification and classification of the test sample. (The graph has been reprinted from Low-cost electrochemical sensor module for measurement of gas concentration© 2016 Grzegorz jasinski, Anna Strzelczyk and Piotr Koscinski, Used under Creative Commons Attribution License (CC BY-NC-SA 3.0). (E) Diagrammatic representation of the working of optical sensors. The apparatus consists of a source of light, it passes the light through the test sample; further, these light rays are sensed by

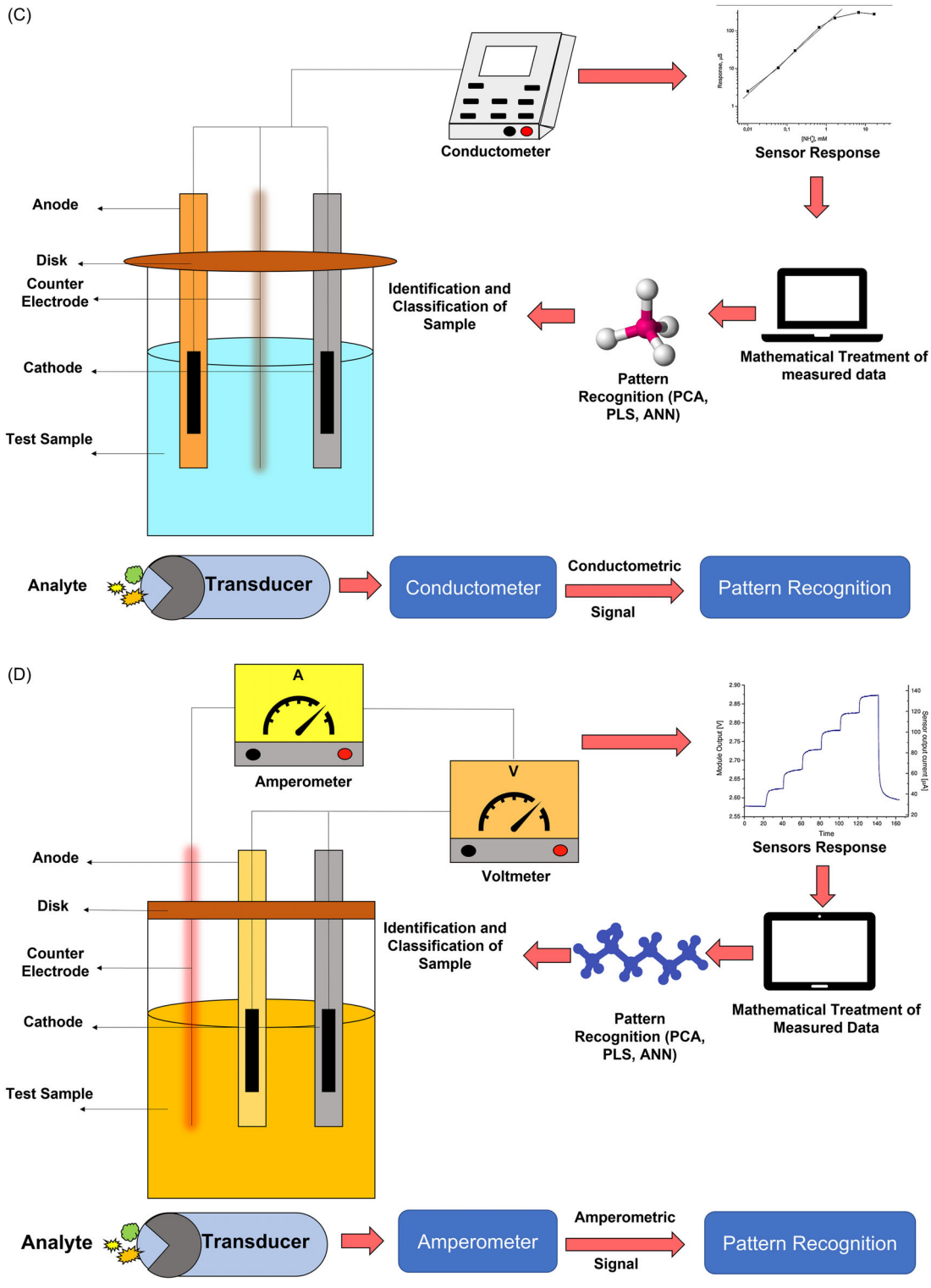


Figure 3. Continued.

optical sensors; these sensors provide the sensor response which provides the pattern recognition (PCA, PLS, ANN techniques) that ultimately leads to the identification and classification of the test sample. (The graph has been reprinted from *Miniaturized optoacoustic feedback sensor for smart laser osteotome: Fiber-coupled Fabry-Pérot etalon sensor*© 2021 Hervé Nguendon Kenhagho; Ferda Canbaz; Raphael Guzman; Philippe Cattin; Azhar Zam, Used under Creative Commons Attribution License (CC BY-NC-SA 4.0). (F) Diagrammatic representation of the working of biosensors. The apparatus consists of three electrodes; namely, anode, cathode, and counter electrode; these electrodes are further connected to the biosensor and are dipped into the test sample; biosensor is connected to the amplifier and this provides the sensor response which further provides the pattern recognition (PCA, PLS, ANN techniques) that ultimately leads to the identification and classification of a test sample. (The graph has been reprinted from *Development of Silicalite/Glucose Oxidase-Based Biosensor and Its Application for Glucose Determination in Juices and Nectars*© 2016 Dudchenko, Oleksandr Ye; Pyeshkova, Viktoriya M.; Soldatkin, Oleksandr O.; Akata, Burcu; Kasap, Berna O; Soldatkin, Alexey P.; Dzyadevych, Sergei V., Used under Creative Commons Attribution License (CC BY-NC-SA 2.0).

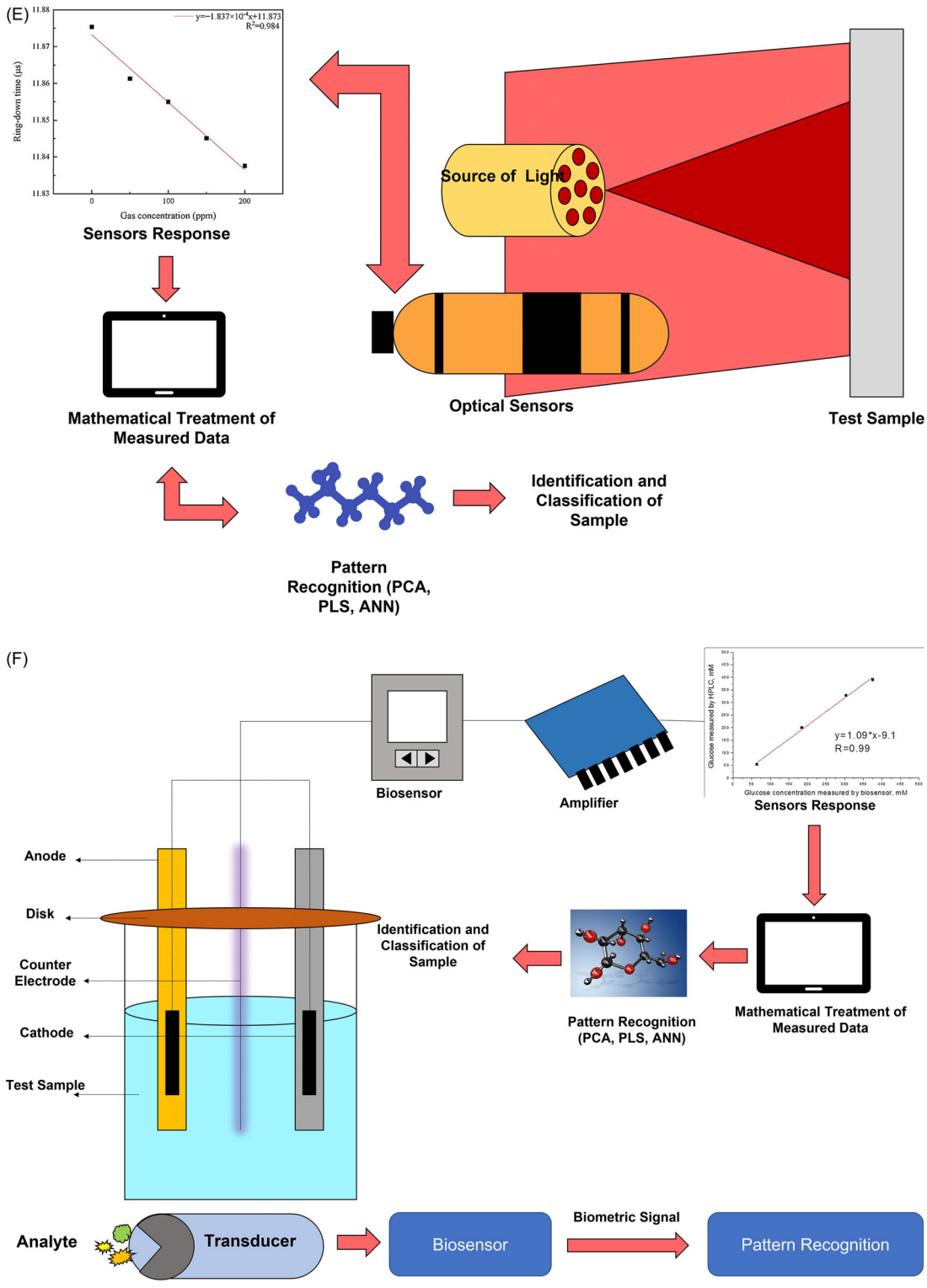


Figure 3. Continued.

resulting in a graph named voltammogram These sensors can be employed for the analysis of various organic and inorganic analytes in various matrices (Blandon-Naranjo et al. 2022).

3.4. Amperometric sensors

It measures the current response to find the analyte concentration at a certain potential. Measurement involves applying a voltage between two electrodes. To prevent harming the electrode, a working potential is only applied briefly. Only when the potential is being applied is the current measured (Yin et al. 2021). During amperometric detection, an electrode experiences an electrochemical conversion (potentiostatically), and the current produced by this electrochemical reaction is then measured as shown in Figure 3(D). Only partial electrolysis occurs when using the amperometric method. The target chemical must be electroactive for this form of detection to work (at the applied potential, in the solution used, and at the prevailing pH). Due to the fact that only

electroactive species may be detected, this has both a limitation and a benefit in that detection can be highly selective (Braz et al. 2022).

The following are the fundamental classes of amperometric sensors used in electronic tongue systems.

3.4.1. Metal sensors

Metals are the most widely used and accessible material for making sensors. Metal electrodes' lack of selectivity is its fundamental shortcoming. It is possible to theoretically oxidize or reduce any chemical whose formal potential is lower or greater than the potential given to the electrode. Electronic tongues that mimic the human taste and are built of metal electrodes are therefore rarely employed alone but rather in combination with other methods. Examples include the ability to anticipate sensory characteristics of Italian red dry wines of various designations of origin using an electronic tongue based on glassy carbon electrodes and a commercial electronic nose (Buratti et al. 2007). Instead of taste assessment, classification applications frequently employ metal electrodes. For this reason, several works have been offered. As working electrodes, five wires made of five different metals—gold, iridium, palladium, platinum, and rhodium—were used in Winquist's description of an electronic tongue. Monitoring was done to see how microbial development was affecting milk quality (Winquist et al. 1998). Winquist gradually added rhenium electrodes, potentiometric sensors, and conductimetric sensors to the pool of earlier metal electrodes. The system was successful in classifying various fermented kinds of milk (Winquist et al. 2000). As demonstrated by Gutes et al. (2005), a variety of metal electrodes, including platinum and carbon (glassy carbon, graphite, etc.), demonstrated a good capacity to distinguish phenolic compounds, which predominate and are characteristic of many food samples. For instance, it has been demonstrated how to categorize wines with various origin designations and those produced in certain geographic regions (Buratti et al. 2004).

3.4.2. Conducting polymer sensors

When polymerized with the right dopant, certain particular monomers exhibit a fluctuation in conductivity with the adsorption of certain analytes. The two conducting polymers that are most frequently used are polypirrole and polyanaline. With the aid of the tools needed for further examination, such polymer films may be quickly and electrochemically manufactured (Vahdatiyekta et al. 2022). A former sensor array that used a microsensor array to distinguish between two analytes was based on polypirrole doped with nine counterions (Nguyen et al. 1999). Subsequently, electronic tongues built from conducting polymers have been used to assess flavours including bitterness, sweetness, acidity, saltiness, and astringency (Apetrei et al. 2004; Arrieta et al. 2004). Recently, a hybrid electronic tongue was employed to distinguish and identify red wines based on their vintage, grape variety, and appellation of origin (Parra et al. 2006). The two primary benefits of conducting polymers are (1) quick analyte adsorption and desorption *via* the film at room temperature, and (2) the ability to establish partial selectivity by changing the polymer's dopant. The primary drawback is that the monomer solution must be distilled before use, and dry solvents are frequently needed to produce a repeatable film.

3.4.3. Biosensors

Amperometric biosensors are made up of two transducers—a solid electrode and an enzyme—that are either in direct touch or in close proximity to one another. Pauliukaite recently put out an example of amperometric biosensors for flavour assessment. An amperometric enzyme-mediator system was used to evaluate umami flavour by measuring the quantity of L-glutamate in tomato-based meals (Pauliukaite et al. 2006). The detection of glucose and ascorbic acid in fruit juices was recently accomplished using a variety of biosensors based on glucose oxidase (GOD) and various metallic catalysts (Gutes et al. 2006). In a batch cell, an array of biosensors based on eight platinum sensors modified with four distinct enzymes was employed to differentiate the quality of wastewater (Tønning et al. 2005).

3.4.4. Nano-composite sensors

Nanomaterials have been shown to significantly increase the performance of metal sensors, conductive polymers, biosensors, and other devices. When one scales down to the nanoscale dimension, the surface-to-volume ratio of the sensors rises, resulting in impressively low detection limits. Fukushima et al. (2020) use nanocomposites and silver nanoparticles to alter the interdigitated electrodes (IDEs) of an impedimetric electronic tongue that was designed to examine taste enhancers and industrial salts. The design of the films that were employed was able to interact with the analytes, allowing for statistical analysis (principal component analysis) to be used to compare and contrast the behaviour of flavour enhancers, commercial salt, and NaCl solutions prepared at various concentrations.

3.5. Optical sensors

They convert light rays into electronic signals. The purpose of an optical sensor is to measure a physical quantity of light and, depending on the type of sensor, then translates it into a form that is readable by an integrated measuring device as shown in Figure 3(E) (Vlasov et al. 2005).

Gutierrez et al. (2010) use a hybrid electronic tongue based on optical and electrochemical microsensors for quality control of wine. The system consists of an array of electrochemical microsensors and a colorimetric optofluidic system. The electrochemical sensor array is made up of two amperometric electrodes, an Au microelectrode, a redox potential sensor, six ISFET-based sensors, a conductivity sensor, and a microelectrode for monitoring electrochemical oxygen demand. A hollow structure, air mirrors, microlenses, and self-aligning structures make up the optofluidic system, which is totally made of polymer technology. A number of multivariate advanced tools have been used to process the data obtained from these sensors, including Principal Component Analysis (PCA) for the pattern recognition and classification of wine samples and Partial-Least Squares (PLS) regression for the quantification of several chemical and optical parameters important for wine quality. The outcomes have shown how useful this technique is for classifying samples into groups based on the grape type and vintage year as well as for measuring a number of sample factors important for wine quality management. Moreover, the optofluidic system and the electrochemical microsensors may be combined on the same platform, resulting in equipment that is quick, efficient, and practical for use when conducting experiments in the field.

3.6. Biosensors

They are devices that produce a measurable signal proportional to the concentration of the target analyte, usually incorporating a biological sensing element and measuring signals derived from biological interactions (Figure 3(F)). Due to their advantageous

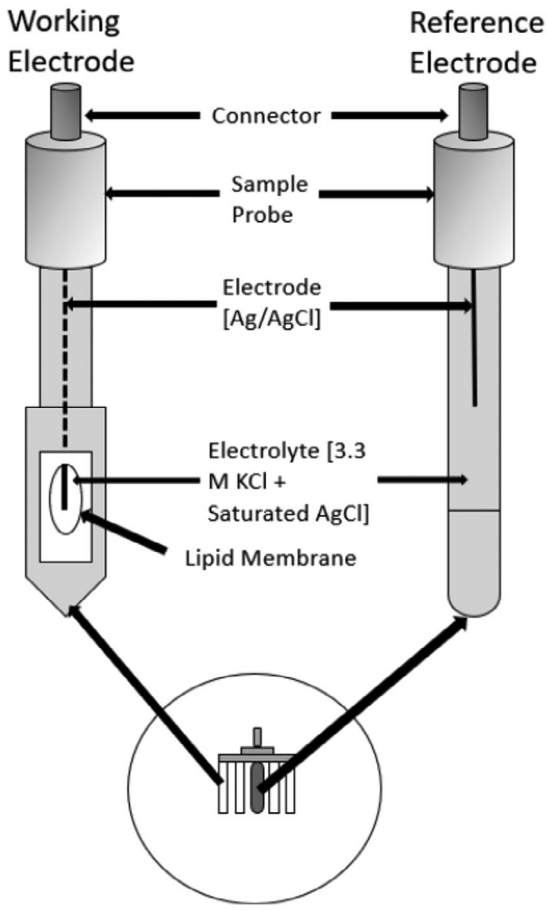


Figure 4. Diagrammatic representation of the working and reference electrode in a sensor. When the reference electrode and sensors were dipped into a beaker containing a test solution, a potentiometric difference between each individually coated sensor with the Ag/AgCl reference electrode was measured, and recorded by the ET software elto data logger.

properties, they have found use in electronic tongue systems used in science, the food industry, and agriculture (Podrażka et al. 2017). Although there are a few examples employing, for instance, microbial sensors or protein receptors, the majority of bioelectronic tongues use enzyme biosensors, which are mostly based on potentiometric or voltammetric/amperometric transduction. Tønning et al. (2005) use the bioelectronic tongue to characterize the type of wastewater using an array of different amperometric biosensors and Gutes et al. (2005) performed the multivariate analysis of a mixture of phenolic compounds from the voltammetric response of a tyrosinase enzyme biosensor. As previously stated, the concept behind the Bioelectronic tongue is to employ biosensors in a multiplex manner, with each biosensor identifying one analyte or a specific collection of analytes, and then using advanced computational algorithms to obtain an accurate characterization of the sample (Del Valle 2017).

Briefly stated, the taste sensing system can be equipped with up to eight lipid membrane sensors, each of which represents a gustatory stimulus or mouth feeling, such as sourness, saltiness, astringency, sweetness, umami, and three types of bitterness specific to molecules with different ionic character. It features a sample table with two parallel circles of sample places. As sensors are coupled to two distinct sensor heads, each sample must be put on both circles in duplicate to prevent cross-contamination. As a result, 10 samples may be measured in a cycle.

The TS-5000Z is the current commercially available system. After immersing the sensors in the sample solution, the change in membrane potential is measured using an Ag/AgCl reference electrode that is independently connected to each sensor head, and mV values are acquired as a result (Figure 4). The sample value is calculated relative to a previously measured reference solution made up of tartaric acid and potassium chloride (Ikezaki et al. 2010). Additionally, following sample readings and two quick cleaning steps (2–3 s), the so-called aftertaste value may be evaluated by submerging sensors in the standard solution. As a result, the substance's adsorption to the membrane is calculated. The obtained value is known as the CPA value (CPA value stands for 'change of membrane potential produced by adsorption'). Sensors are cleaned in an ethanolic solution before measuring the next sample, which takes place 30 s after the previous sample and aftertaste measurements, respectively. The provider pre-sets the measurement period to 30 s by default, although it is manually adjustable (Zhang et al. 2022).

4. General sample preparation and e-tongue operational conditions

The compounds tested were weighed and dissolved in purified water. All testing beakers contained 80–100 ml of solution. When the reference electrode and sensors were dipped into a beaker containing a test solution, a potentiometric difference between each individually coated sensor with the Ag/AgCl reference electrode was measured and recorded by the ET software elto data logger. Each sample was analyzed for 120 s (Poghossian et al. 2019). The liquid sensors and the reference electrode were then rinsed with purified water for 10 s after each sample analysis. Using well-conditioned sensors, each sample was usually tested eight times by the rotation procedure (i.e. the first round of measurements of all samples was completed before the next round of measurements started. The potentiometric contrast made between every individual sensor and the reference terminal was estimated and recorded by the e-tongue BPM programming. All examples were examined at room temperature (Magro et al. 2019).

5. How long does measuring take?

Usually, 40 min is taken per sample to measure as the sensor stabilization and washing take up 90% of the overall time. The equipment is unable to measure samples of neutral chemicals. Electric potential difference sensors and reference electrodes are read by the sensor. The idea is related to the workings of human taste perception (Riul et al. 2003; Lozano-Torres et al. 2022).

6. Sensory evaluation

According to various experts, profile analysis is one of the most complicated descriptive sensory evaluation methodologies. As a result, we employed it in our investigations. This technique is divided into three major parts. The panelists first develop the vocabulary of descriptors based on the test samples. In the second step, with the assistance of the panel leader, each sensory attribute must be properly described and understood by all panelists. The panel selects an evaluation technique after defining the sensory qualities (e.g. unstructured scale, category scale, etc.). During the third phase, the panelists work independently in sensory booths to analyze samples according to previously defined methods. The goal of these tests is to offer an accurate portrayal

of samples. The findings characterize the investigated materials from a sensory standpoint. One significant advantage of this approach is that it allows for a comparison of different samples based on perceived tastes. The consensus group established and rated the following features on unstructured scales: 'hue', 'transparency', 'bubbles', 'sweet odor', 'sour odor', 'apple odor', 'off odor', 'sweet taste', 'sour taste', 'apple taste', 'off taste', and 'taste persistence'. ProfiSens software creates electronic data test sheets, distributes them to panelists, and collects the completed data sheets (Woertz et al. 2010)

Consumers and the ET capacity for discrimination between samples of paneer cheese with various amounts of capsaicin (1.875, 3.75, 7.5, 15, and 30 ppm) were assessed by Schlossareck and Ross. Consumers ($n = 110$) assessed samples over the course of two days while adopting a blocked design to reduce tiredness. At 3.75, 7.5, 15, and 30 ppm (P 0.05), customers could tell the spicy paneer sample apart from the control (0 ppm). Between samples 3, 7, and 15 ppm, differences were discovered ($p < 0.05$). However, no appreciable variations were discovered between the control and 1.875 ppm or between samples 15 and 30 ppm. Although both high and low concentrations presented difficulties for consumers, the e-tongue produced a high level of discrimination of 93% across all samples. The spicy, sweet, salty, sour, and umami sensors were connected to PC1, which accounted for 85.6% of the variability on the principal component analysis plot produced by the e-tongue. PC2 accounted for 7.96% and was connected to metallic and bitter senses. PC1 specified the 3.75 and 15 ppm samples, whereas PC2 defined the 1.875 and 30 ppm samples. These findings suggest that in addition to sensory evaluation, the e-tongue may be effective for qualitative paneer differentiation (Woertz et al. 2010)

7. Cross-sensitivity test

Five compounds were used for the cross-sensitivity test including tartaric acid (sourness), sodium saccharin (sweetness), quinine (bitterness), sodium chloride (saltiness), and monosodium glutamate (umami). A concentration of 10 mM of Tartaric acid, sodium saccharin, sodium chloride, and monosodium glutamate were used while quinine was made at 1 mM (Wang et al. 2022).

7.1. Case study

A report in 2014 from food safety regulators of the Barabanki district of Uttar Pradesh mentioned that there are higher levels of monosodium glutamate (MSG) which is used as a taste enhancer apart from high lead content which is above the permissible level and on the labeling of Maggi noodles, it was mentioned that there is no added MSG. It was sent to a state laboratory in Gorakhpur for testing, it was found that it contains MSG. After a year in 2015, the Gorakhpur lab report was confirmed by the central food laboratory (CFL) and it was stated that the amount of lead was above 1000 times than Nestle India Ltd. Claimed (Rajain et al. 2019).

8. Multivariate data analysis

Multivariate data analysis is utilized to assess the working electrode's ability to categorize the samples. An artificial neural network (ANN) is utilized for quantitative analysis, whereas the PCA technique is applied to the qualitative analysis of the data set. For the PCA analysis, the data sets for each electrode have been taken into account (Janczyk et al. 2010; Vlasov et al. 2005). Prior to PCA, each variable is mean-centered and scaled to unit variance as part of the data analysis process, which also incorporates an initial pre-processing step employing an auto-scaling technique on the transient responses. Baseline subtraction and auto-scale techniques, respectively, have been used for pre-processing in the cases of backpropagation multilayer perceptron (BP-MLP) and radial basis function (RBF) neural network classifiers to reduce the variability of the input signal while preserving the relevant information (Vlasov et al. 2005).

PCA has been used to test the behavior of the changed and unmodified electrodes in identifying the sample's floral type. A huge number of observations, or the current responses of various floral varieties to each electrode, were used to create the data set. To highlight variation and pattern within a data collection, PCA is applied to each data set. As it provides a visual representation of the experimental data for each type of sample and electrode, the PCA plot is crucial. The data were processed by the ANN in order to create a classifier model. The network model may accurately approximate any function (Guedes et al. 2021).

9. Applications

Numerous industrial sectors, including the pharmaceutical and food and beverage industries, have used electronic tongues in various ways (Figure 5). It is applicable for:

9.1. Bitterness evaluation

Different formulations, both with and without medicines (placebo), have had their bitterness evaluated. The Alpha Mos Electronic Tongue has been used to assess the stability of taste and odor in placebo formulations using various pharmaceutical ingredients (Wang et al. 2022). By matching bitter placebos, good identification of the various doses of additives was also achieved with excellent reproducibility. On multiple samples aged using various aging techniques, the successful separation of the distinct scents, as well as the monitoring of taste degradation, were obtained. Data processing and identification methods included Principal Component Analysis (PCA), Discrimination Function Analysis (DFA), and Partial Least Square (PLS) for quantification. By displaying data for various formulations (syrups, oral dispersive, gels, etc.) and diverse medication types, the principle of qualitative measurements with the electronic tongue was discovered (Ciosek and Wróblewski 2007). Most medications have a naturally bitter taste, which can cause localized tongue numbness or a burning sensation in the throat. The effectiveness of the masking agents used to lessen the bitterness of medications must therefore be evaluated. Finding a medicine formulation that has the bitter taste masked requires comparing many placebo and drug formulations (Del Valle 2012).

9.2. Pharmaceutical formulation design

The effectiveness of various formulations, such as tablets, syrups, powders, capsules, lozenges, effervescent, emulsions, sublingual, orodispersible, films, etc., at disguising taste is measured by ET (Woertz et al. 2010). It aids in the selection of the top flavour-masking substances. Improve the effectiveness of the active (Nutraceutical) formulation's flavour masking even before the safety assessment investigation. It evaluates the effect of each ingredient's flavour. Improve formulation design and choice by measuring the bitterness intensity. Increase your selection of substances without consistently using a sensory panel. In one of the

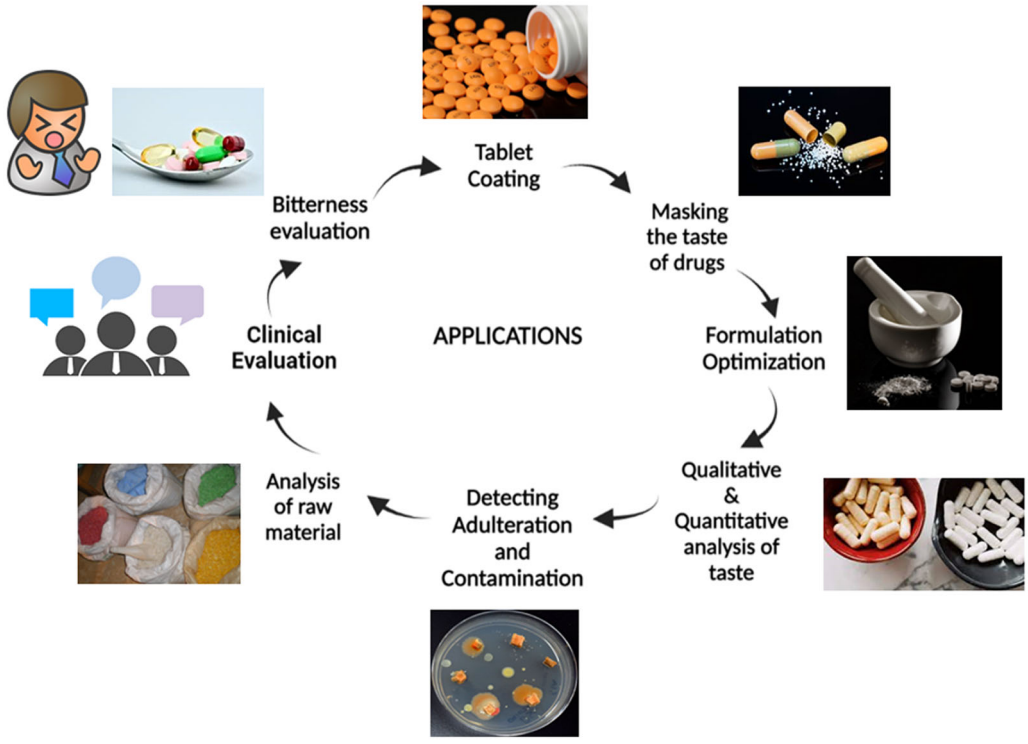


Figure 5. Diagrammatic representation of the applications of the electronic tongue. The primary function of the tongue in humans is to distinguish between various flavours. Because of this, flavour evaluation and the research for substances that affect taste perception are the most frequent uses of its artificial equivalent. Even if all of the component parts were measured, it could still be impossible to determine the real flavour of a food product because of a synergistic or suppressive impact.

investigations, the effectiveness of taste-masking was assessed using quinine as a model bitter chemical and the sweetener acesulfame as a bitterness inhibitor (Ikezaki et al. 2010). The findings imply that the sensors may be used to optimize the sweetener level in a liquid formulation by detecting the inhibition of bitterness by a sweetener. The ET could also detect the bitterness inhibition of quinine when it was combined with other well-known taste-masking excipients, such as sodium acetate, sodium chloride, HPMC, polyvinyl alcohol, and ethyl cellulose (Munekata et al. 2023).

9.3. Masking the taste of drugs

The use of Active Pharmaceutical Ingredients (API) including paracetamol, ibuprofen, diclofenac, caffeine, and many others is essential for the effective treatment of infections and disorders. The ability to taste distinguishes between a variety of tastes, each of which is naturally connected with either pleasure or distaste. Unfortunately, API flavour is typically described as bitter and is hence seen to be unpleasant (Koichi et al. 2009). The method which is frequently used involves covering tablets and capsules with a sugar film or a coating that has no flavour. Unfortunately, due to swallowing issues, liquid versions of medications are sometimes the only ones that can be used in juvenile and geriatric therapy. Several sorts of taste masking strategies are employed to avoid or lessen API's bitterness, including the inclusion of sweets and artificial tastes. Initially, panels of willing human participants were asked to explain their preferences and thoughts. Laboratory animals, particularly mice, and rats, can partially replace human testing panels. However, it is ethically questionable to use human test subjects and lab animals to evaluate the flavour of medications in their early stages of research because allergies may result in fatalities. Additionally, findings from human test panellists differ amongst people because of the high degree of taste complicity, and in the event of small tests, groups only offer subjective data. Age, sex, and the tester's country of origin can all affect how they perceive flavours (Woertz et al. 2010). The electronic tongue was designed to address the requirement for an artificial instrument for taste judgment that complies with ethical norms. Two commercial systems (Alpha MOS Astreell, Toulouse, France, and Intelligent Sensor Technology Inc., Atsugi-shi, Kanagawa, Japan, TS-5000Z) devoted to the analysis of API taste masking have been developed as a result of studies on a correlation between mice and/or human test panels and an electronic tongue. The development of new sensor types that can assess API flavour masking techniques is ongoing. In order to distinguish between common APIs like ibuprofen and roxithromycin, a potentiometric sensor array was used, and the results demonstrated that microencapsulation had a comparable impact on the taste masking of both substances. Newman et al. suggested conducting research on the bitter flavour of sodium caseinate hydrolysates, which are a rich source of bioactive peptides, and the use of sweets as masking agents (Schlossareck and Ross 2019).

9.4. Formulation optimization

Drug and bitterness-mimicking compositions can have their bitterness measured accurately by ET. It enables quick comparison to accepted references in under 2 min. Time and money are saved, which leads to an improvement in product quality, purity, and consumer acceptance. It will be easier to establish consistent taste masking if the taste and flavour characteristics of medicinal formulations are controlled from the preformulation stage to full-scale manufacture. It is possible to keep track of product consistency across several locations, geographical areas, or even contract manufacturers. Process engineers and product development

experts may now monitor the flavour quality of medicine formulations due to the availability of this new technology (Guedes et al. 2021; Yin et al. 2021).

9.5. Qualitative and quantitative assessment of taste

The ET offers a qualitative and quantitative analysis of taste judgment early in the development process. The risk to humans is decreased while generating speedy results to advance test compounds to the following research stage by decreasing the necessity to recruit human taste test subjects. It swiftly assesses the many formulation candidates so that the formulator may choose the ideal mixture of chemicals (additives) and medications that produce the 'best'-tasting formulation (Aouadi et al. 2020).

9.6. Evaluates the chemical stability of substances (flavours, ingredients, additives) or completed goods (drugs, food, and beverages)

The stability of the medication in terms of taste and the detection of medicinal scents are also analyzed. Evaluate the evolution of flavour and the consistency of oral formulations in relation to different storage scenarios, latency, or manufacturing. Compare distinct formulations' stability profiles effectively under stress or as a result of age (Lu et al. 2016).

9.7. Analysis of raw materials

This approach can be used to examine raw materials or finished goods for uniformity of quality/origin, storage stability, storage effectiveness, and contaminant recognition (Blandon-Naranjo et al. 2022).

9.8. Clinical evaluation

Recognizing and quantitatively analyzing both organic and inorganic components in biological and pharmaceutical fluids disease diagnosis *via* a comprehensive chemical study of liquid media. ET has been used to analyze medicinal fluids with a similar composition to blood plasma. The system has the ability to conduct simultaneous multicomponent analysis, measuring the concentration of a number of components, including some that are extremely challenging to measure using other techniques, such as hydrocarbonate and phosphate. ET has been used to differentiate between urine samples from healthy people and patients with various illnesses. The method can identify between urine samples from healthy individuals and those that include blood as a result of various illnesses. Additionally, it is utilized for *in vivo* clinical monitoring (Ciosek and Wróblewski 2007).

9.9. Detecting adulteration and contamination

It is not only illegal but also potentially dangerous for the consumer's health if there are undesired substances present that are not listed on the label. Therefore, it is crucial to develop quick, accurate, and affordable technologies to identify food contamination (Munekata et al. 2023).

Recently, a genetically modified acetylocholinesterade-based automated hydrodynamic bioelectronic tongue was used to quantify pesticide (chlorpyriphos-oxon and malaoxon) combinations in milk. In a different investigation, the measurement of formaldehyde, urea, and melamine in milk was made possible using a voltammetric electronic tongue with detection limits (10.0, 4.16, and 0.95 mmol/L, respectively), which were below the upper range of the advised tolerated consumption level (Podrażka et al. 2017).

10. Limitations of electronic tongue

1. The system's sensors can only recognize a standard set of tastes that have already been entered into the database (Munekata et al. 2023).
2. The sensors are capable of accurately identifying the flavour of the sample, but they are unable to perfectly replicate the sophisticated human olfactory system. Similar to human senses, artificial senses cannot be integrated simultaneously (Schlossareck and Ross 2019).
3. The device's durability is also a problem because of the sensors it uses, which shortens its lifespan. Additionally, some of the market's sensors are expensive (Del Valle 2012).
4. As long as a sample is liquid, it can be measured. We often dissolve API or the medication being tested in 10mM KCl solution to give it some conductivity. Purified water is typically used to dissolve or combine with food samples during food testing to remove the flavour. Oil should be removed to less than 2% for the best sensor conditions (Anjiki et al. 2011).

11. Future scope of electronic tongue

As medical technology advances, it is anticipated that sales of ET will expand at a significant rate. Over the course of the projected period, rising research and development for the production of novel drugs is anticipated to fuel demand for ET (Ghasemi-Varnamkhasti et al. 2018). It is anticipated that the increased use of cutting-edge technologies and equipment in pharmaceutical manufacturing facilities will boost sales. Given that they allow pharmaceutical scientists to do studies without having to worry about the safety of humans, their demand in the pharmaceutical sector is projected to grow significantly over time (Podrażka et al. 2017). Increasing government support for the innovation of medical diagnosis and medication development producers is anticipated to promote market expansion. Increasing innovative devices and sensor development for taste evaluation of dissolved organic and inorganic substances support the increase in demand for ET (Ahmad et al. 2006). Future market trends are anticipated to benefit from efforts to cut costs along the value chain in order to sell products at prices that are competitive with the market. Additionally, long-term performance depends on advancing manufacturing, sustaining innovation, and customer-focused business practices. Simplified the sensory test process for calculating the bitterness of drugs. These features have increased the demand for ET significantly (Del Valle 2012).

12. Conclusion

In modern chemical sensor research, the electronic tongue is a growing and promising subject. Global selectivity is a notion that it uses. It is an effective method for evaluating and forecasting the flavour of medications and associated products. In regular pharmaceutical development and manufacturing analysis, it takes the place of human panels. The exposure and danger associated with conducting human panel testing for goods are significantly decreased, allowing for superior analytical outcomes to swiftly determine the optimum formulation and launch the product.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The author(s) reported there is no funding associated with the work featured in this article.

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